

Chapter 10

Fractal Geometry and the Limits of Prediction

This chapter studies examples in which simple recursive rules produce surprisingly complex behavior. Some recursive processes lead to regular growth patterns that are easy to describe. In other cases, repeated application of a rule creates behavior that is difficult to predict over the long term.

We will look at several examples: the Collatz process, self-similar fractals such as the Koch curve, and the logistic map. Together, these examples show that simple rules can lead either to regular structure or to highly complicated behavior. The chapter closes by connecting these ideas to the practical limits of prediction.

10.1 Recursion Revisited

The two recursions introduced in Chapter 1 have appeared in every subsequent chapter. The additive recursion $a_{n+1} = a_n + d$ produced linear growth: savings accounts under simple interest, cost and revenue functions, the perimeter of the Koch curve. The multiplicative recursion $a_{n+1} = r \cdot a_n$ produced exponential growth: compound interest, the number of filled triangles in the Sierpinski construction, the doubling of bacteria populations.

Both of these recursions are **predictable**: given any starting value and the rule, the value at step n can be computed exactly without stepping through all intermediate values.

Type	Rule	Value at step n
Additive	$a_{n+1} = a_n + d$	$a_0 + nd$
Multiplicative	$a_{n+1} = r \cdot a_n$	$a_0 \cdot r^n$

Not all recursions have this property. A recursion whose long-term behavior depends sensitively on the starting value, so that small changes in the starting value produce large differences in the outcome, is said to exhibit **sensitive dependence on initial conditions**. This is the defining feature of chaotic systems.

Example 10.1.1. Compare two starting values in the additive recursion $a_{n+1} = a_n + 3$:

Starting value $a_0 = 10$: produces 10, 13, 16, 19, ...

Starting value $a_0 = 10.001$: produces 10.001, 13.001, 16.001, ...

The two sequences remain close together. The difference at step n is always 0.001. This is a **stable** recursion.

Example 10.1.2. Compare the same two starting values in the recursion $a_{n+1} = 2a_n$:

$a_0 = 1.0$: 1, 2, 4, 8, 16, 32, 64, 128, ...

$a_0 = 1.001$: 1.001, 2.002, 4.004, 8.008, 16.016, 32.032, 64.064, 128.128, ...

The relative difference remains constant at 0.1%. The absolute difference grows (from 0.001 to 0.128 after seven steps) but proportionally the sequences track each other. This recursion is **predictably divergent** but not chaotic.

Example 10.1.3. Now consider the recursion $a_{n+1} = 4a_n(1 - a_n)$, starting from two nearby values.

$a_0 = 0.500$:

$$a_1 = 4(0.500)(0.500) = 1.000$$

$$a_2 = 4(1.000)(0.000) = 0.000$$

$$a_3 = 0.000, a_4 = 0.000, \dots$$

The sequence reaches 0 and stays there.

$a_0 = 0.501$:

$$a_1 = 4(0.501)(0.499) = 1.000 - 0.004 = 0.9980$$

$$a_2 = 4(0.998)(0.002) = 0.0080$$

$$a_3 = 4(0.008)(0.992) = 0.0317$$

$$a_4 = 4(0.0317)(0.968) = 0.1228$$

$$a_5 = 4(0.1228)(0.877) = 0.4308$$

The sequence moves through $(0, 1)$ in an apparently irregular way.

A starting value of 0.500 produced a sequence that collapsed to zero in two steps. A starting value of 0.501 produced a sequence that wandered unpredictably. The difference in starting values was 0.001. This is sensitive dependence on initial conditions.

Exercises 10.1

10.1.1. Apply the additive recursion $a_{n+1} = a_n + 5$ starting from $a_0 = 10$ and from $a_0 = 10.1$. Compute 5 steps for each.

- (a) What is the difference between the two sequences at each step?
- (b) Does this difference grow, shrink, or stay constant?
- (c) Is this recursion sensitive to initial conditions?

10.1.2. Apply the recursion $a_{n+1} = 3a_n$ starting from $a_0 = 1.0$ and from $a_0 = 1.01$. Compute 6 steps for each.

- (a) Compute the absolute difference at each step.
- (b) Compute the relative (percentage) difference at each step.
- (c) Does the relative difference change?

10.1.3. The recursion $a_{n+1} = 4a_n(1 - a_n)$ is applied starting from $a_0 = 0.3$. Compute 8 steps. Then apply the same recursion starting from $a_0 = 0.31$.

- (a) At what step do the two sequences first differ by more than 0.1?
- (b) What does this tell you about predicting long-term behavior from an approximate starting value?

10.2 The Collatz Conjecture

Definition. *Collatz recursion.* The **Collatz recursion** is defined by two rules applied according to whether the current value is even or odd:

$$a_{n+1} = \begin{cases} a_n/2 & \text{if } a_n \text{ is even,} \\ 3a_n + 1 & \text{if } a_n \text{ is odd.} \end{cases}$$

The **Collatz conjecture** states that for every positive integer starting value, the sequence eventually reaches 1. Once 1 is reached, the sequence cycles: $1, 4, 2, 1, 4, 2, \dots$

This conjecture has been verified computationally for every starting value up to approximately 2.95×10^{20} . It has not been proven for all positive integers. The celebrated mathematician Paul Erdős said of it: “Mathematics is not yet ready for such problems.”

Example 10.2.1. Apply the Collatz recursion starting from $a_0 = 6$.

$$6 \rightarrow 3 \rightarrow 10 \rightarrow 5 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1.$$

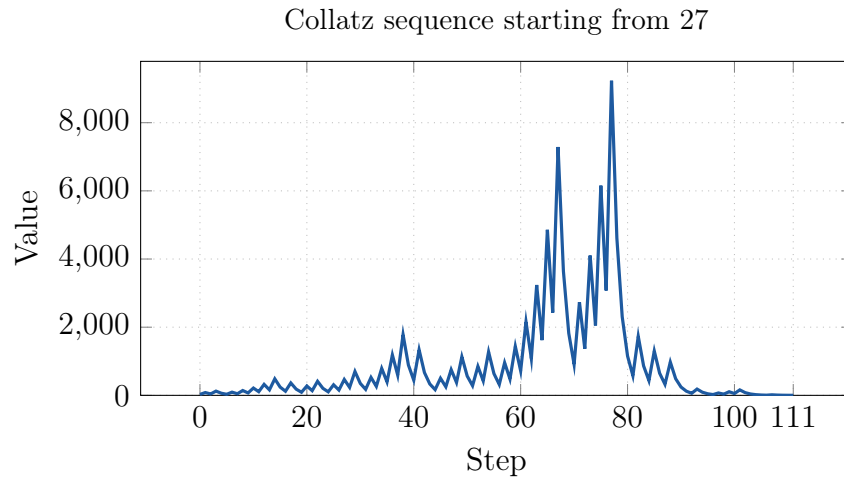
The sequence reaches 1 in 8 steps.

Example 10.2.2. Apply the Collatz recursion starting from $a_0 = 27$. The sequence takes 111 steps to reach 1 and reaches a peak value of 9,232.

Selected values from the trajectory:

$$27 \rightarrow 82 \rightarrow 41 \rightarrow 124 \rightarrow 62 \rightarrow 31 \rightarrow 94 \rightarrow 47 \rightarrow 142 \rightarrow \dots \rightarrow 9232 \rightarrow \dots \rightarrow 1.$$

The sequence rises dramatically, falls, rises again, and eventually reaches 1. There is no simple formula predicting how long this will take or how high the sequence will go.



Example 10.2.3. (*Comparing nearby starting values*) Starting values 26 and 27 differ by 1.

Starting from 26: $26 \rightarrow 13 \rightarrow 40 \rightarrow 20 \rightarrow 10 \rightarrow 5 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$. Reaches 1 in 10 steps. Peak value: 40.

Starting from 27: reaches 1 in 111 steps. Peak value: 9,232.

Two starting values separated by 1 produce trajectories of lengths 10 and 111 with peak values 40 and 9,232. This is sensitive dependence on initial conditions in a particularly stark form.

Exercises 10.2

10.2.1. Apply the Collatz recursion to each starting value. Record each step until the sequence reaches 1. Count the number of steps.

- (a) $a_0 = 5$
- (b) $a_0 = 11$
- (c) $a_0 = 16$
- (d) $a_0 = 24$

10.2.2. Compare the Collatz sequences starting from $a_0 = 6$ and $a_0 = 7$.

- (a) Record the full sequence for each.
- (b) How many steps does each take to reach 1?
- (c) What is the peak value of each sequence?
- (d) What does the difference in trajectories illustrate about the Collatz recursion?

10.2.3. The Collatz recursion uses both division (additive reduction) and multiplication (multiplicative growth) depending on whether the current value is even or odd. In two complete sentences, explain how this connects to the two recursions studied throughout this course.

10.2.4. The Collatz conjecture has been verified for all starting values up to approximately 2.95×10^{20} but has not been proven for all positive integers. In two complete sentences, explain why computational verification for a very large finite set of cases does not constitute a mathematical proof.

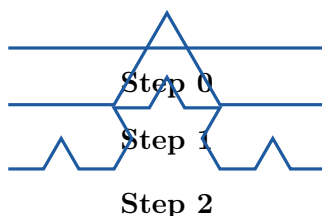
10.3 Self-Similarity and Fractal Geometry

The Sierpinski triangle and Koch curve are examples of **deterministic fractals**: shapes produced by applying a fixed rule repeatedly and exactly at every scale. Their self-similarity is perfect, meaning that the same pattern reappears under magnification.

Many natural phenomena exhibit approximate self-similarity across scales, but not the perfect regularity of a deterministic construction. Coastlines, mountain ranges, river networks, and bronchial passages all show similar patterns at different scales without being exactly self-similar. These are studied using the tools of fractal geometry.

The Koch Curve and Fractal Dimension

The Koch curve is constructed by the rule introduced in the course summary. At each step, every segment is replaced by four segments of one-third the length.



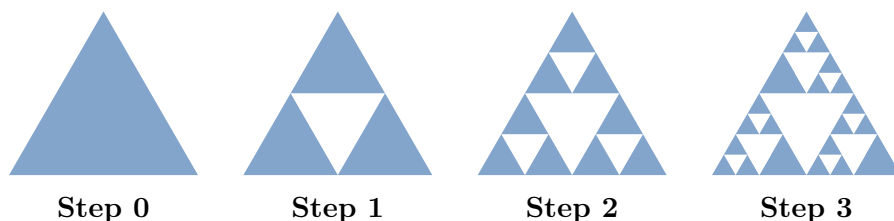
Step	Segments	Length of each segment
0	1	1
1	4	$1/3$
2	16	$1/9$
3	64	$1/27$
n	4^n	$(1/3)^n$

The total length at step n is $4^n \times (1/3)^n = (4/3)^n$. Since $4/3 > 1$, this grows without bound. The Koch curve has infinite length.

The Koch curve has dimension $\log_3 4 \approx 1.262$, a value between 1 (a line) and 2 (a filled region). Fractal dimension measures how completely a curve fills the plane as measurement scale decreases.

The Sierpinski Triangle

The Sierpinski triangle is constructed by removing the central triangle from a filled equilateral triangle, then applying the same rule to each remaining triangle.



Step	Filled triangles	Fraction of area remaining
0	1	1
1	3	$(3/4)^1$
2	9	$(3/4)^2$
3	27	$(3/4)^3$
n	3^n	$(3/4)^n$

The fraction of area remaining converges to zero. The Sierpinski triangle contains infinitely many points arranged in a self-similar pattern, yet has area zero. Its fractal dimension is $\log_2 3 \approx 1.585$.

Exercises 10.3

10.3.1. Fill in the missing values in the Koch curve table.

Step	Segments	Length of each	Total length
0	1	1	1
1	4	$1/3$	$4/3$
2			
3			
4			

- (a) Is the total length growing according to an additive or multiplicative recursion?
- (b) What is the ratio of total length from one step to the next?

10.3.2. Fill in the Sierpinski triangle area table.

Step	Filled triangles	Fraction of area
0	1	1
1	3	$3/4$
2		
3		
4		

- (a) The filled triangle count follows a _____ recursion.
- (b) The fraction of area remaining follows a _____ recursion.
- (c) At step 10, what fraction of the original area remains? Compute $(3/4)^{10}$ and round to four decimal places.

10.3.3. A new fractal is constructed from a segment of length 1 as follows: divide into five equal parts and replace the middle part with three sides of a square. The resulting curve has 7 segments each of length $1/5$.

- (a) Fill in the table for steps 0 through 4.
- (b) What is the ratio of total length from one step to the next?
- (c) Does this curve have finite or infinite length? Justify.

10.4 The Logistic Map

The Koch curve and Sierpinski triangle showed that a simple deterministic rule can produce objects of infinite geometric complexity. The logistic map shows that a simple deterministic rule can produce behavior that, over time, is statistically indistinguishable from randomness.

Definition. *Logistic map.* The **logistic map** is the recursion:

$$x_{n+1} = r \cdot x_n(1 - x_n),$$

where $x_n \in (0, 1)$ and $r \in (0, 4]$ is a fixed parameter.

The logistic map was originally introduced as a model of population growth with a carrying capacity. If x_n represents the population as a fraction of the maximum sustainable level, then x_{n+1} gives the population in the next generation. The term $(1 - x_n)$ acts as a brake: when the population is near its maximum, growth slows.

The behavior of the logistic map depends critically on the value of r .

Example 10.4.1. (*Stable fixed point: $r = 2.5$*) Starting from $x_0 = 0.5$:

$$\begin{aligned}x_1 &= 2.5(0.5)(0.5) = 0.625 \\x_2 &= 2.5(0.625)(0.375) = 0.586 \\x_3 &= 2.5(0.586)(0.414) = 0.606 \\x_4 &= 2.5(0.606)(0.394) = 0.597 \\x_5 &= 2.5(0.597)(0.403) = 0.601\end{aligned}$$

The sequence converges to a single value near 0.6. This is called a **fixed point**: a value x^* such that $r \cdot x^*(1 - x^*) = x^*$. For $r = 2.5$, the fixed point is $x^* = 1 - 1/r = 1 - 0.4 = 0.6$.

Example 10.4.2. (*Period-2 oscillation: $r = 3.2$*) Starting from $x_0 = 0.5$:

$$\begin{aligned}x_1 &= 3.2(0.5)(0.5) = 0.800 \\x_2 &= 3.2(0.8)(0.2) = 0.512 \\x_3 &= 3.2(0.512)(0.488) = 0.799 \\x_4 &= 3.2(0.799)(0.201) = 0.514 \\x_5 &= 3.2(0.514)(0.486) = 0.799\end{aligned}$$

The sequence oscillates between approximately 0.513 and 0.799. This is called a **2-cycle**: the sequence repeats every two steps.

Example 10.4.3. (*Chaotic behavior: $r = 4.0$*) Starting from $x_0 = 0.5$:

$$\begin{aligned}x_1 &= 4.0(0.5)(0.5) = 1.000 \\x_2 &= 4.0(1.0)(0.0) = 0.000 \\x_3 &= 0.000\end{aligned}$$

The sequence collapses to zero.

Starting from $x_0 = 0.501$:

$$x_1 \approx 0.999$$

$$x_2 \approx 0.004$$

$$x_3 \approx 0.016$$

$$x_4 \approx 0.062$$

$$x_5 \approx 0.234$$

$$x_6 \approx 0.717$$

$$x_7 \approx 0.811$$

Starting values 0.500 and 0.501 produce completely different trajectories after two steps. At $r = 4$, the logistic map is fully chaotic.

Value of r	Behavior	Long-run pattern
$0 < r < 1$	Decay to zero	Single value: 0
$1 < r < 3$	Stable fixed point	Single value: $1 - 1/r$
$3 < r < 3.45$	Period-2 cycle	Alternates between 2 values
$3.45 < r < 3.54$	Period-4 cycle	Alternates between 4 values
$r \rightarrow 3.57$	Increasing periods	Cycles of length 8, 16, ...
$3.57 < r \leq 4$	Chaos	Unpredictable

The transition from stable to chaotic behavior as r increases is called **period doubling**. The lengths of the stable cycles double at each transition: 1, 2, 4, 8, 16, ... This is the multiplicative recursion from Chapter 1, now describing the structure of the path to chaos.

In the chaotic regime, the long-run distribution of values of x_n follows a probability distribution over $(0, 1)$. A process that is entirely deterministic, with no randomness in its rule, produces output that looks like a sample from a probability distribution. The connection to Chapter 2 is precise: what appeared to be random variation may, in some systems, be the output of a deterministic but chaotic rule.

Exercises 10.4

10.4.1. Apply the logistic map $x_{n+1} = r \cdot x_n(1 - x_n)$ for each value of r and starting value. Compute 8 steps and describe the qualitative behavior.

(a) $r = 2.0$, $x_0 = 0.4$

(b) $r = 3.0$, $x_0 = 0.5$

(c) $r = 3.5$, $x_0 = 0.5$

10.4.2. For $r = 2.8$, the fixed point is $x^* = 1 - 1/r$.

(a) Compute x^* .

(b) Starting from $x_0 = 0.2$, apply the logistic map for 10 steps.

(c) Does the sequence appear to converge to x^* ?

10.4.3. The logistic map models population dynamics. Suppose x_n is the population of fish in a lake as a fraction of the maximum sustainable population.

(a) For $r = 1.5$, what happens to the fish population in the long run?

(b) For $r = 3.2$, the population oscillates between two values. Interpret this in terms of fish population.

(c) For $r = 3.9$, the population behaves chaotically. What does this imply for a fisheries manager trying to set annual catch limits? Answer in two complete sentences.

10.5 Fractals as Constraint Boundaries

Chapters 8 and 9 assumed that feasible regions have smooth, straight-line boundaries. Each constraint produced a line segment as a boundary, and the feasible region was a polygon with a finite number of vertices. Real constraints, however, are sometimes described by irregular or fractal boundaries.

The Coastline Paradox

Suppose a city must confine development to land within 10 miles of the coastline. The feasible region for development is bounded on one side by the coastline.

If the coastline is a straight line, the feasible region is a strip with a well-defined area. If the coastline is irregular, the boundary becomes more complex. If the coastline is fractal, the boundary has infinite length, and the area of the feasible region depends on how finely the boundary is resolved.

Definition. *Coastline paradox.* The **coastline paradox** is the observation that the measured length of a coastline increases as the measurement scale decreases. A shorter ruler captures more of the small-scale irregularities and produces a larger measurement.

This was first analyzed mathematically by Benoit Mandelbrot in 1967. The coastline of Great Britain, measured at different scales, produces lengths ranging from approximately 2,800 km (measured with a 200 km ruler) to over 17,800 km (measured with a 1 km ruler).

Example 10.5.1. A surveyor measures a coastline using a ruler of length ℓ . At each step, the ruler is halved. The number of ruler placements grows as $n(\ell) \sim \ell^{-D}$, where D is the fractal dimension of the coastline.

For the coast of Great Britain, $D \approx 1.25$. This means the measured length grows as $L(\ell) = n(\ell) \cdot \ell \sim \ell^{1-D} = \ell^{-0.25}$.

As $\ell \rightarrow 0$, $\ell^{-0.25} \rightarrow \infty$. The length grows without bound as measurement precision increases.

The coastline paradox connects to the feasible region analysis of Chapters 8 and 9 in a direct way. A feasible region bounded by a fractal curve has no well-defined perimeter. The area of such a region can be finite even when the boundary has infinite length. This is the same relationship between area and perimeter seen in the Koch snowflake: a curve of infinite length enclosing a finite area.

Example 10.5.2. (*Fractal-bounded feasible region*) A city planner defines a conservation zone as the set of all points within distance d of a river whose path is approximated by a Koch curve of step n .

Step n	River segments	Approximate perimeter of zone
0	1	proportional to $(4/3)^0 = 1$
1	4	proportional to $(4/3)^1 \approx 1.33$
2	16	proportional to $(4/3)^2 \approx 1.78$
3	64	proportional to $(4/3)^3 \approx 2.37$

The perimeter of the zone grows without bound as finer detail is included. The area of the zone, however, converges to a finite value as n increases. Two planners using different map scales will measure different perimeters for the same conservation zone. The area, being less sensitive to scale, is a more reliable description of the zone's extent.

Exercises 10.5

10.5.1. A surveyor measures a coastline with rulers of length 100, 50, 25, and 12.5 meters. The number of ruler placements are 80, 200, 520, and 1,340 respectively.

- Compute the measured length for each ruler.
- Does the measured length increase or decrease as the ruler gets shorter?
- What does this imply about defining the “true length” of the coastline?

10.5.2. A rectangular feasible region has vertices at $(0, 0)$, $(10, 0)$, $(10, 5)$, $(0, 5)$ and a perimeter of 30 units. Its area is 50 square units. Now replace the top edge with a Koch curve of step 2.

- (a) The top edge at step 0 has length 10. What is the length of the top edge at step 2 of the Koch construction?
- (b) Does replacing the top edge with a Koch curve change the area of the region significantly? Explain in one sentence.
- (c) What difficulty would arise in applying the linear programming method of Chapter 9 to a feasible region with a fractal boundary?

10.5.3. In two complete sentences, explain why the area of a fractal-bounded region can be finite even when the boundary has infinite length. Use the Koch snowflake as your example.

10.6 The Recursion Thread: A Final View

Every chapter of this course has been organized around two recursions. The additive recursion $a_{n+1} = a_n + d$ produced steady, linear growth: number line steps, simple interest, cost functions, constraint boundaries. The multiplicative recursion $a_{n+1} = r \cdot a_n$ produced accelerating growth: compound interest, binary place values, the segment count in the Koch curve.

This chapter has shown that recursion is not limited to these two patterns. The logistic map $x_{n+1} = r \cdot x_n(1 - x_n)$ combines additive and multiplicative elements and produces behavior that depends on r in a complex way: stable, periodic, and chaotic behavior all arise from the same rule. The Collatz recursion uses different rules for even and odd values and produces trajectories whose lengths and peak values defy simple prediction.

Chapter	Additive recursion	Multiplicative recursion
1	0, 3, 6, 9, . . .	1, 2, 4, 8, . . .
2	Frequency counts	Relative frequencies
3	Steps from the mean	Squared deviations
4	Confidence interval width	Normal distribution
5	Decimal positions	Powers of 2 (binary)
6	Simple interest	Compound interest
7	Fixed cost, fixed revenue	Profit per additional unit
8	Constraint boundaries	Feasible region corners
9	Objective function values	Optimal vertex search
10	Perimeter of Koch curve	Area of Sierpinski triangle

The two recursions were present from the first page. They are present on this one. The course has been, in one sense, a ten-chapter exploration of what two simple rules can produce when applied in different contexts and at different scales.

The Limits of Observation

An observer with perfect knowledge of a stable system can predict its future exactly. An observer measuring a periodic system can predict its future exactly after identifying the period. An observer measuring a chaotic system cannot predict its long-term future, no matter how precisely the present state is measured, because small measurement errors are amplified exponentially by the recursion.

This is the final and most complete statement of the theme that has run through every chapter. In Chapter 2, the question the observer asks determines which feature of the distribution is visible. In Chapter 4, the resolution of the sample determines the width of the confidence interval. In Chapter 8, the observer's choice of objective determines which vertex is optimal. In Chapter 10, the precision of the initial measurement determines how far into the future the observer can see.

Mathematics describes both what can be known and what cannot. The tools developed in this course are instruments for extending the range of the observer. They do not eliminate the limits of observation. They locate those limits precisely.

Exercises 10.6

10.6.1. Identify the chapter in this course where each of the following ideas first appeared. Then explain in one sentence how Chapter 10 extends or completes that idea.

- (a) The multiplicative recursion.
- (b) The observer's question determines the answer.
- (c) The same rule applied at different scales.
- (d) Probability distributions arising from data.

10.6.2. The Collatz conjecture, the logistic map, and the Koch curve are all generated by simple recursive rules. In three complete sentences, describe what is similar and what is different about the behavior produced by each rule.

10.6.3. In the logistic map at $r = 4$, two starting values $x_0 = 0.5$ and $x_0 = 0.500001$ will eventually produce completely different trajectories. In two complete sentences, explain why this makes the logistic map unsuitable as a predictive model of population growth when the initial population can only be measured approximately.

10.6.4. Write a paragraph of four to six sentences connecting the tools of this course to the theme of observation. Your paragraph should reference at least three chapters and explain how each tool extends or limits what an observer can know.

Chapter 10 Review

6. Apply the Collatz recursion to each starting value. Count the number of steps to reach 1 and record the peak value.

- (a) $a_0 = 18$
- (b) $a_0 = 19$
- (c) Do starting values 18 and 19 (which differ by 1) produce trajectories of similar or different length and peak?

6. Fill in the Koch curve table through step 5.

Step	Segments	Total length
0	1	1
1	4	$4/3$
2		
3		
4		
5		

Does the total length converge to a finite value or grow without bound?

6. Apply the logistic map $x_{n+1} = 3.8x_n(1 - x_n)$ starting from $x_0 = 0.4$. Compute 10 steps.
 - (a) Does the sequence appear to settle into a fixed point, a cycle, or chaotic behavior?
 - (b) Apply the same map starting from $x_0 = 0.41$. At what step do the two sequences first differ by more than 0.05?
6. A coastline is measured with rulers of length $\ell = 10, 5, 2.5$, and 1.25 km. The respective number of ruler placements are 50, 130, 340, and 900.
 - (a) Compute the measured length for each ruler.
 - (b) Plot the four (ruler length, measured length) pairs on a graph.
 - (c) Describe the trend.
6. A feasible region in a linear program has straight line boundaries. Describe what would change about the linear programming method from Chapter 9 if one boundary were replaced by a Koch curve of step n for large n . Address the concepts of vertices, feasibility checking, and the Corner Point Principle.
6. Explain in three complete sentences how the behavior of the logistic map connects to the probability distributions studied in Chapter 2. Your explanation should address: what the distribution represents, why it arises from a deterministic rule, and what this implies about interpreting data that appears random.
6. The theme of this course has been that what an observer sees depends on the question being asked and the scale at which it is asked. Write a paragraph of five to seven sentences tracing this theme through at least five chapters. For each chapter you

mention, name the specific mathematical tool and explain what it reveals and what it conceals about the underlying situation.